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DEPARTMENT OF INFORMATION TECHNOLOGY

**“ Court information management system for Tepi town
yeki wereda”**

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Abstract

The efficient management of court information is crucial for ensuring justice and maintaining transparency in legal proceedings. Traditional manual court record-keeping systems present numerous challenges, including data loss, inefficiencies in case management, and limited accessibility to case information. This research focuses on developing a web-based Court Information Management System (CIMS) for Tepi Town, Yeki Wereda.

The purpose of this research is to design and implement a system that digitizes and streamlines the handling of both civil and criminal cases, reducing redundancy, improving security, and facilitating faster case processing. The methodology employed includes data collection through interviews and document analysis, system design using object-oriented modeling, and system development using MySQL, PHP, and JavaScript. The system was evaluated based on its usability, performance, and security.

The results indicate that the proposed CIMS significantly enhances court efficiency by reducing manual workload, improving case tracking, and ensuring data security. Stakeholders reported increased accessibility and a more transparent judicial process.

In conclusion, the implementation of a digital CIMS is a crucial step toward modernizing court operations, reducing costs, and improving service delivery. It is recommended that further enhancements, such as integration with higher court systems and online judgment features, be considered for future development.

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Acronyms and abbreviation

PC.....	Personal computer
HTML.....	Hypertext markup language
CSS.....	cascading stayle sheet
PHP.....	Hypertext Preprocessor
PHP.....	Hypertext Preprocessor
JS.....	javascript
RAM.....	Random accesses memory
CMS.....	Court Management System
OOA.....	Object-oriented analysis
OOD.....	Object-oriented design
UML.....	Unified model language

1.1 Background of the Organization

- ❖ The court office was established during the "Derg" regime. Before this time disagreement and criminal cases were solved in traditional way. Starting from that time it passes many obstacles to reach today's position. The court office of tepi town yeki wereda established in 1996 E.C started to serve the customer in modern ways. The organization is located around the Tepi town administration on the right side and on the left side of old police station in hibret kebele 02. The purpose of the court is to provide legal decision for those plaintiffs and defendants based on their evidences as well as for those who commit crimes. The lawyers make decision based on the constitution of the country.
- ❖ The court information management system addresses two major cases those are civil case and criminal cases. It has the responsibility to support, manage, giving response to the case, taking case from the kebele (mahiberawifirdeshengo) to wereda level court office.
- ❖ We have seen some problems in data processing and handling. During the process there are many difficulties of manual processing of files/data. Not only this but also giving comments for one court is difficult because of distance. Now we are going to develop software that can solve problems of data processing, data handling and etc.

1.2 Background of the Project

- ❖ The system based on the court information management system that improves the manual data processing. Court Information Management System is document management system or handling of data, dataflow system, appointment and also concern with a customers' comment. There are two major types of cases. Those are civil law and criminal law. Each of them contains their own sub branches. In order to execute those cases some process takes place. As we visited the work place of how the court information management system takes place. we have seen some problems in data processing in tepi town and handling. During the process there may be many difficulties of manual processing of files/data. Not only this but also giving comment for one court may be difficult because of distance. This

means there is no online giving comment service before for the court found on the distance. Now we are going to develop software that can solve problems of data processing, data handling, give appointment ,assign case ,registering case and view assigned case etc. we can achieve our aim by using hardware (pc) and software (programming language).

- ❖ The main purpose of doing this project is that to save person's file from damage, store files forever, secure files that must be secured, reduce costs and time. There will be data communication from one office to the other. There will be many advantages after the end of our project.

1.3 Statement of the problem

- ❖ Court information management system of tepi town yeki wereda does on both criminal and civil case type. As mentioned above these major types contain so many sub branches. Criminal case is initiated when crime is committed, then investigation held to collect information and evidence about what is happened to determine who committed the crime. After that the accused is charged with the crime and taken into custody. Based on evidence, then the judge determine whether he/she guilty or not guilty. At the end the decision is presented. This process is done manually.
- ❖ We have seen the following problems as follows
 - ✓ Difficulty of storing the documents neatly
 - ✓ Files may be stolen by thieves, robbers or internal attackers.
 - ✓ Files may be fade because of long life
 - ✓ Files may be destroyed by natural disasters like fire , flood
 - ✓ Customer can't see some news on distance.
 - ✓ Difficulty of data store in secure way.
 - ✓ Customer can't inform appointments day and time remotely or from every where.
 - ✓ Recording examination, cross examination and reexamination may difficult while collecting evidence from witness.
 - ✓ Difficult to giving appointments for customers.

- ✓ Difficult to evaluate every judge decisions.
- ✓ Difficult to assign cases to judges

1.4 Objective

1.4.1 General objective

- ❖ Our main objective is to develop web based court information management system for tepi town yeki wereda .

1.4.2 Specific objectives

- ❖ This system includes Specific Objective:
 - ✓ Analyze the existing system and design better system for the organization
 - ✓ To design a way to solve the problem.
 - ✓ To design and developed a user friendly system to handle data insertion, updating, deletion, retrieving on the database.
 - ✓ To develop a secured Data base system.
 - ✓ To develop suitable system to give appointment and assign cases.
 - ✓ To develop the court information management system.

1.5 Literature Review

The field of court information management has undergone significant transformations due to the advancements in information technology. Several studies highlight the inefficiencies of manual systems in legal institutions, emphasizing issues such as mismanagement of case files, loss of documents, and delays in justice administration (Walker, 1980). The shift towards digital court management systems has been observed in various jurisdictions to improve operational efficiency and ensure secure data handling (Jacobson, 2003).

Court Information Management Systems (CIMS) have been adopted globally to automate case handling and document management processes. These systems leverage databases and web technologies to enhance accessibility and security. For instance, Cockburn (2006) discusses the benefits of integrating essential use cases in legal information systems to facilitate seamless information flow between different court departments.

The major functionalities of modern court management systems include case registration, document storage, scheduling, and real-time access to legal information. Studies indicate that implementing a web-based court management system significantly reduces the risks associated with manual record-keeping, such as data breaches, unauthorized access, and physical deterioration of case files. Additionally, digital solutions allow for better tracking of case progress, ensuring that legal procedures are followed efficiently (Jacobson, 2003).

Despite the advantages of automated systems, challenges such as limited technological infrastructure, user adaptation, and data security concerns remain prevalent. Addressing these issues requires a well-structured system design that incorporates robust security measures, user-friendly interfaces, and scalable architectures (Cockburn, 2006). The proposed web-based CIMS for Tepi Town, Yeki Wereda, aligns with these best practices by incorporating secure authentication, efficient data retrieval, and role-based access control.

In conclusion, the literature suggests that the adoption of digital court management systems enhances legal service delivery by improving accessibility, security, and efficiency. The development of the CIMS for Tepi Town aims to address the limitations of manual processes and contribute to the broader effort of judicial digital transformation. This study builds upon existing research by customizing a web-based solution to meet the specific needs of the local judicial system, ensuring a more streamlined and effective approach to court case management.

1.6 Methodology of our project

1.6.1 Data Sources

- ❖ Both primary and secondary sources of data are used to gather information.

1.6.1.1 Primary Data Source

- ✓ By interviewing respective personal of the workers on the court office.
- ✓ Collect end users feedback about the existing system from the customers and judges.

1.6.1.2 Secondary Data Source

- ✓ Relevant documents and techniques that helps to improve the performance of website Just after data collection requirement analysis is done and the next stage of the project is system design, and object oriented design is used to design the system so that the output will enable to visualize the objects in the system in a simple and understandable form.

1.6.2 Data gathering Techniques

- ❖ Since data gathering is the process of gathering and measuring information on different area of the system.
- ❖ We will implement the gathering technique in the following ways :
- ✓ We observe how the site works.
- ✓ We observe how the site really needed to be managed and modified.
- ✓ We observe how data 's are not updated frequently on the site.

1.6.3 Systems Analysis and Design

- ❖ It is the process of breakdown an entire system into module, analyzing each module separately and determining the relationship between the system development process we used system modeling. i.e. creating model of the system.
- ❖ System modeling is the process of creating a model of system by analyzing and organizing the system elements. This is the phase where deeply understanding of the existing system problems and finding alternative solution. To design the system, we choose the concept of Object-oriented system analysis and design (OOSAD) Object-oriented modeling techniques and Unified modeling language (UML) tools.
- ❖ object oriented modeling technique is advantages for the following reasons why we use it:
 - ✓ It enables us to comprehensively model a system before we develop it.
 - ✓ Modification of the object implementation is easy because objects are loosely coupled.
 - ✓ Understanding of the structure is easy because object-oriented modeling and tools used to represent real world entities.
 - ✓ Direct manipulation of architectural components is possible because several object-oriented programming languages exist.
 - ✓ UML is also the main tool we used for necessary software design and analysis.

- ✓ To develop diagrams and provide users with ready-to-use expressive modelling examples.
- ✓ It can be used for modelling a system independent of a platform language.

1.6.4 Overview of the Project

- ❖ The new system will include solutions for the existing system problems by giving services better than the older one as much as possible. After observing the current manual court information management system and evaluating all the problems that occurred during every activity on the existing system, the project team has desired to design a web-based court information management system for the tepi town yeki wereda with the following improvements:-
 - ✓ **Performance:-** the system will have a faster response time and use minimal space usage.
 - ✓ **Speed:-** regarding the speed the system will generate output within a short time.
 - ✓ **Reliability:-** the system is reliable by analyzing each business rule set by the organization
 - ✓ **Security:-** the system provides or contains a user name and password for each user based on their privilege this increases the security of workflow.
 - ✓ For administrative it is best to see the weekly, monthly, and yearly reports and activities. It also helps to know how the judge is resolving problems and it helps to get feedback from the clients about the court information management system. For the organization, the system will help save cost and time.
 - ✓ The new proposed system is used to minimize the number of tasks labeled to the employees.
 - ✓ **Economy:-** when we apply a Web-based court information management system there will be areas in which cost will be reduced. Example as a result of a new system the payment too many employees will be reduced. Reduce the cost of paper and other materials.
 - ✓ **Efficiency:-** the Web-based court information management System by itself is short and clear and in this system there is no duplication of data through the new system is powerful to manage things.

1.6.5 Development Tools and technology

1.6.5.1 Software Tools

- ❖ The following requirement software tools are used to implement our project work.

Front end technology

- ✓ **HTML:** used for user interface design
- ✓ **CSS:** used for styling HTML user interface.
- ✓ **JavaScript:** Makes it interactive
- ✓ **Bootstrap:** reponsive design mobile friendly site
- ✓ **Browser:** Google chrome , Mozilla Firefox

Back-end technology

- ✓ **Server:** Apache HTTP Server is used run the web server within Windows OS.
- ✓ **Database:** MySQL is common database in creating dynamic websites it is a high-speed database.
- ✓ **PHP:** is a scripting language that can be used to access data from the database.
- ❖ Generally, Apache, MySQL, and PHP are open-source components that can be installed individually, they are usually installed together.

NO	Activity	Tools / programs
1	HTML, CSS, JS	Client-side programming
2	PHP	Server-side scripting
3	MYSQL	For database
4	MS Power point	For presentation
5	Draw.io	to draw UML diagram
6	Browser	Google chrome , Mozilla Firefox
7	MS windows	Platform
8	Microsoft word	to prepare documentation
9	XAMPP	For database

Table 1: Software tools

1.6.5.2 Hardware Tools

NO	Hardware name	Description
1	Printer	For printing out design prototype and document in hard copy
2	Computer	To do computing process
3	Internal/external hard disk	For data storage and backup
4	LCD projector	For presentation

Table 2 : hardware tools

1.7 Scope of the project

- ❖ The scope of our project is to implement a web-based court information management system, which is aimed to avoid the problems associated with the manual system which focuses on tepi town.
- ✓ Accurate way of recording and storing information into the database.
- ✓ Administrator can control everything and assign roles to each of them as they see fit.
- ✓ Make the system high secure, efficient and effective
- ✓ View assigned cases that case distributor are assign to Judges.
- ✓ The assigned judges will set decide judgement day.
- ✓ To be able to notify customers of the judgement day by message when the judges assigned.
- ✓ Register any customer that wants to open new case.
- ✓ President and vice president view reports.
- ✓ avoiding the data redundancy of the system.
- ✓ Search the files in short times from many files.
- ✓ Update employee's information.
- ✓ Accept comment that customers are writing for court offices

1.8 Feasibility

1.8.2 Operational Feasibility

- ❖ The system requires very little human power. It provides a simple user interface in which the employee of the office can access easily, and more accurately. Maintenance is less and easy. The system will have GUI and very little user training is required to learn it. The system is so simple to use, so no special skills will be required to use the system. So the system is operationally feasible.

1.8.3 Technical Feasibility

- ❖ The system is to be developed by using technologically system development techniques such as PHP, Java script, CSS, and MySQL database without any problems and the group members have enough capability to develop the project. So the system will be technically feasible.

1.8.4 Behavioral /Political feasibility

- ❖ The political feasibility is a feasibility that makes sure the system does not break law. Our system does not violent with any law, hence the system is politically feasible.

1.8.5 Economic Feasibility

- ❖ The system to be developed is economically feasible and the benefit is out weighing the cost. Since this project already computerizes the existing system, now the reduction of cost for materials used in manual operation becomes beneficial to the organization.

1.9 Limitation of the project

- ✓ The system does not send appeals to the high court and Supreme Court because there is no interconnected web-based system with zonal court office.
- ✓ Our system has no online judgment
- ✓ Our system has no sound and video record.

1.10 Significance of the project

- ❖ After the implementation of this project, it provides various benefits. as follow
- ✓ Can save their financial budget.
- ✓ Customers will give their comment online without going there.
- ✓ Customer can see some news on distance.
- ✓ Customer informed appointments day and time remotely..
- ✓ Adequate and fast communication with judge and other coworker.
- ✓ Assigning new case to judge easy and secure more with this this system.
- ✓ The judges find appointment date and time easily for many case in short period of time.
- ✓ Recording examination, cross-examination, and reexamination is easy while collecting.
- ✓ It doesn't require any extra hardware materials like paper, pen, pencil and etc.
- ✓ It reduces the wastage of time during searching and modifying employee
- ✓ Reduce cost paid for extra hardware materials like pen, paper, and printing.
- ✓ Decrease Error in filing the client document.
- ✓ Information about court can be accessed any time from any place based on given privilege.

1.11 Task and Schedule

NO	Activity	Start	Finish	Nov 17	Dec 17	Jan 17	Feb 17	Mar 17	Apr 17	Mar 17	Jun 17
1	Proposal	25/03/17	02/04/17								
2	System analysis	03/04/17	04/05/17								
3	System Design	06/05/17	01/06/17								
4	Implementation	02/06/17	03/09/17								
5	Testing	04/09/17	02/10/17								

Table 3: Task and Schedule

1.12 Cost analysis of the project

NO	Material Required	Amount	Price per unit	Total prices
1	Flash	3	350	1,050 birr
2	Personal computer	3	30,000	90,000 birr
3	Note book	3	120	360 birr
4	pen	5	30	150 birr
5	paper	1	500	500 birr
Total				92,060 birr

Table 4: cost of the project

1.13 Communication plan

Communication plan describes the schedule of group member or the time of communication plan to develop project. The group member meets four days in each week.

day	Time	Place (class room)
Monday	Local(4:00-6:00)pm	LAB
Tuesday	Local(3:00-4:00)	LAB
Thursday	Local(3:00-5:20)pm	LAB
Saturday	Local(2:00-5:00)am	LAB

Table 5: communication plan

1.14 Project Team Composition

NO	NANE	IDNUMBER	Responsibility
1	HAMID TAREKE	MTUUR/1601/14	In all activity
2	EBENEZER SYUM	MTUUR/1148/14	In all activity
3	ABNET GEREMEW	MTUUR/0170/14	In all activity
4	ABREHAM AMANUEL	MTUUR/08/14	In all activity
5	MIHRET FIKRU	MTUUR/2256/14	In all activity

Table 6: project team composition

1.15 Conclusion

- ❖ The development and implementation of a web-based Court Information Management System (CIMS) for Tepi Town, Yeki Wereda, demonstrate significant improvements in court operations. The system successfully addresses the inefficiencies of manual processes by digitizing case management, enhancing data security, and providing real-time accessibility to case information. By leveraging modern technologies, the proposed system facilitates efficient court operations, minimizes paperwork, and reduces administrative burden. The findings indicate that adopting digital solutions in the judicial sector significantly improves case tracking, decision-making, and overall efficiency in court administration.

1.16 Recommendation

- ❖ According to the scope of our project, the project team develops a web based system for tepi town court office. Because of the time constraint, we cannot do beyond to scope, but in the future the team believes that this system can be fully operational by having enough time and full information. Neatly, the team would recommend that further work should be done on the system.
 - **Integration with Higher Courts:** Expanding the system to integrate with regional and national court databases for seamless data exchange and case appeals processing.
 - **Online Case Filing Services:** Enabling litigants to file cases online and access judgment reports remotely to enhance accessibility and convenience.
 - **User Training and Capacity Building:** Providing training sessions for court staff and stakeholders to ensure effective utilization and maintenance of the system.

1.17 References

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1.18 Appendix

- ❖ Here are the lists of questions that we were used to interview and data gathering with the Tepi town yeki wereda court office.
 1. how many sectors are there according to the current office strategies and work situation?
 - 2, How the sectors and employees work their activity?
 3. how we could to get any manual data from the information store for example Annual schedule, List of sectors etc.
 4. Who schedule the annual and semi-annual schedule?
 5. What is the role of law officer, judge, advocator public proctor client president?
 6. When the report is generated?
 7. How much max duration of one case to complete?
 8. How does your current system work?
 9. How you make relationship with the user?
 10. What are the procedures or steps when customers get service of the organization?
 11. How do you store customer's data and other information?
 12. What are the problems that you face when the users use the existed system?